Essential Clinical Symptoms & Signs

KEMRI | Wellcome Trust

saving children's lives
ETAT+
Emergency Triage Assessement and Treatment plus admission

University of Nairobi

KENYA PAEDIATRIC ASSOCIATION

Objectives

- Define key symptoms of common, serious diseases
- Define and demonstrate key clinical signs of common, serious diseases
- Improve communication between professionals

The most useful symptoms & signs

- Observed commonly in common illnesses
- Help assessment of nature and severity of illness
  - Indicate risk of death
  - Useful for monitoring progress
  - Differentiate diseases
- Easy for everyone to observe and learn

Choosing the ‘best’ symptoms & signs

- WHO and others investigating this for 30 years
- Sound evidence base for most common disorders of children
- Included in the IMNCI approach
- Best signs are the foundation of the whole week

Symptoms 3

Length of illness	5 days	
Fever – No. of days 3	Y☑	N☐
Cough– No. of days 5	Y☐	N☐
Cough > 2 weeks	Y☐	N☑

Why is it important to document the duration of fever? of cough?

Cough for more than TWO weeks is not acute pneumonia

Contact with TB /Chronic cough (last 12 months)	Y☐	N☐
Difficulty breathing	Y☐	N☐

Why is history of contact with TB /chronic cough important.

What does ‘contact’ mean?

Why ‘last 12months’?

Symptoms 4

Diarrhoea No. of days =	Y☐	N☐
Diarrhoea > 14d	Y☐	N☐
Diarrhoea bloody	Y☐	N☐

Why do we need to ask whether there is diarrhoea >14 days or whether it is bloody?

Vomiting, No / 24hrs =	Y☐	N☐
Vomits everything	Y☐	N☐
Difficulty feeding	Y☐	N☐

Vomiting everything means no oral medicines and is a danger sign

Convulsions: No. = 2	Y☐	N☐
Partial / focal fits?	Y☐	N☐

Convulsion >1 *or* Partial convulsions suggest meningitis – a danger sign. Requires LP

Airway / Breathing

- Airway
  - Stridor (inspiratory)

- Breathing adequacy
  - Respiratory Rate – Counted for 1 minute in a calm child!
  - Oxygen saturation (pulse oximetry)
  - Central cyanosis
  - Head nodding
  - Grunting
  - Indrawing
  - Acidotic / deep breathing
  - Wheeze / crackles

Central Cyanosis

- Gums / Tongue
- NOT fingers
- Lips unreliable
- Problem detecting cyanosis if the child has severe anaemia

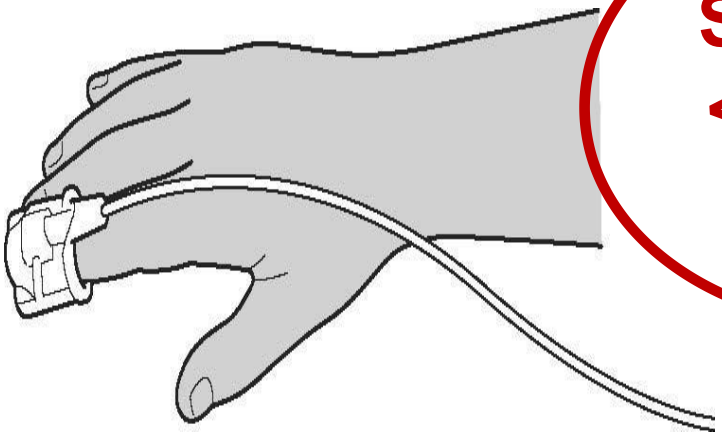
Indrawing

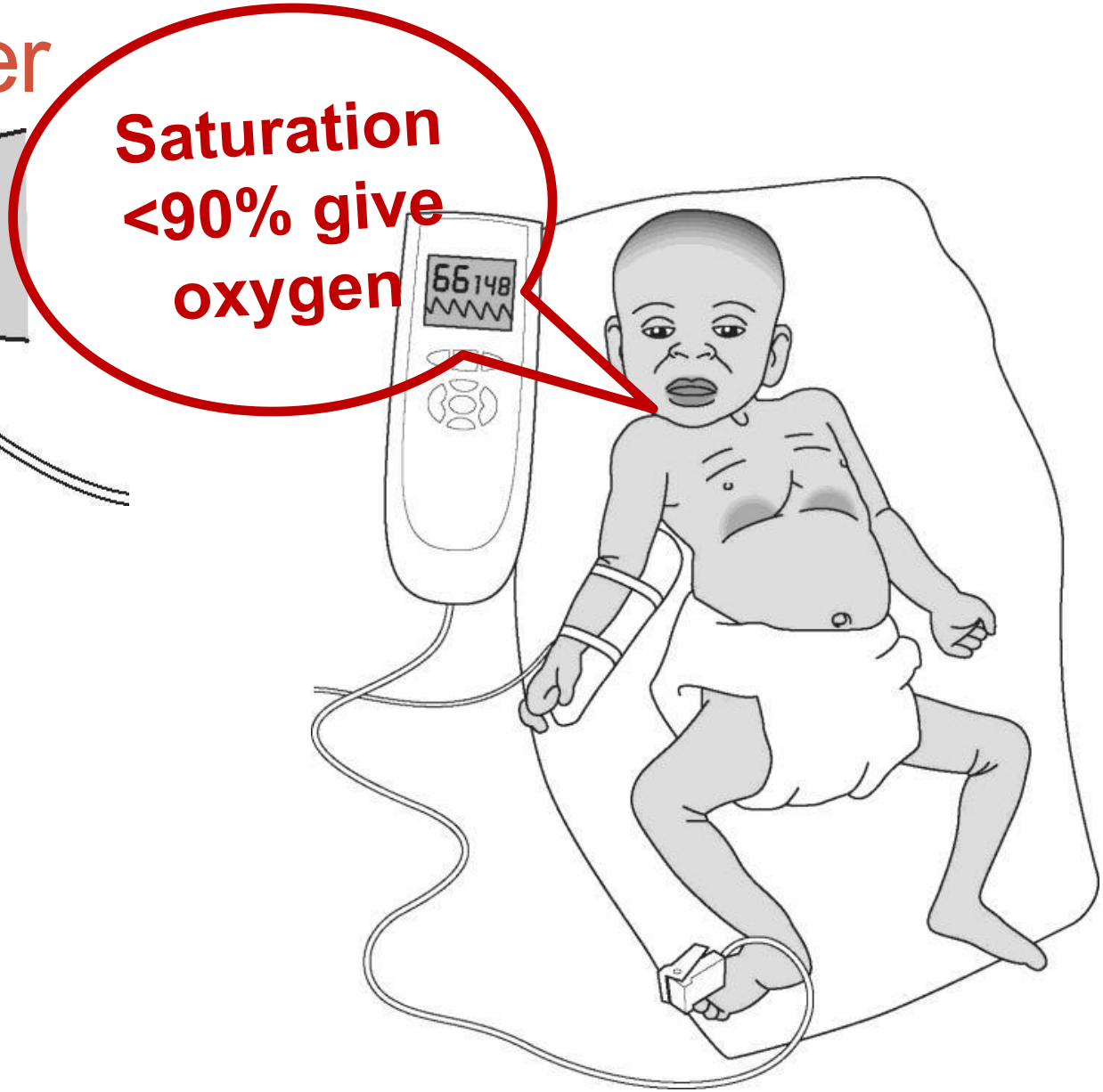
Deep/Acidotic Breathing

Respiratory distress

- Signs
  - Grunting:
    - abnormal, short, deep, hoarse sounds on exhalation
  - Grunting is the body's way of trying to keep air in the lungs so they will stay open

Pulse oximeter





Signs 2 – Circulation & Dehydration

Peripheral Pulse	☐ Normal	☐ Weak	
Cap Refill	secs	*X = not possible*	
Pallor / Anaemia	**0**	**+**	**+++**
Skin warm at:	☐Hand	☐Elbow	☐Shoulder
Sunken eyes	**Y**☐	**N**☐	
Skin pinch (sec)	**0**	**1**	**≥ 2**

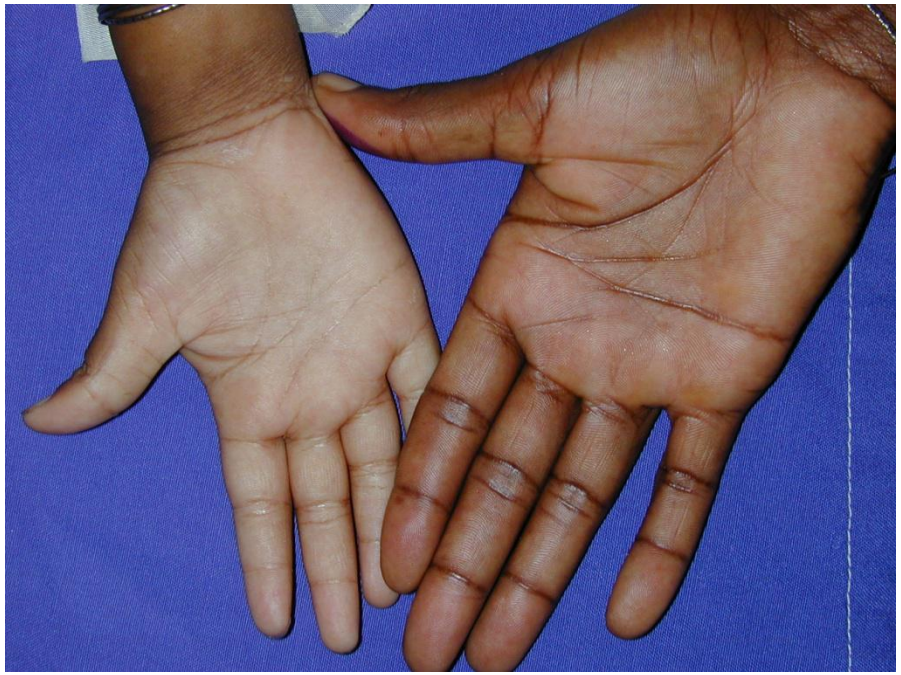
- Pulse
  - Weak (or absent)
  - Rate

Capilllary refill time

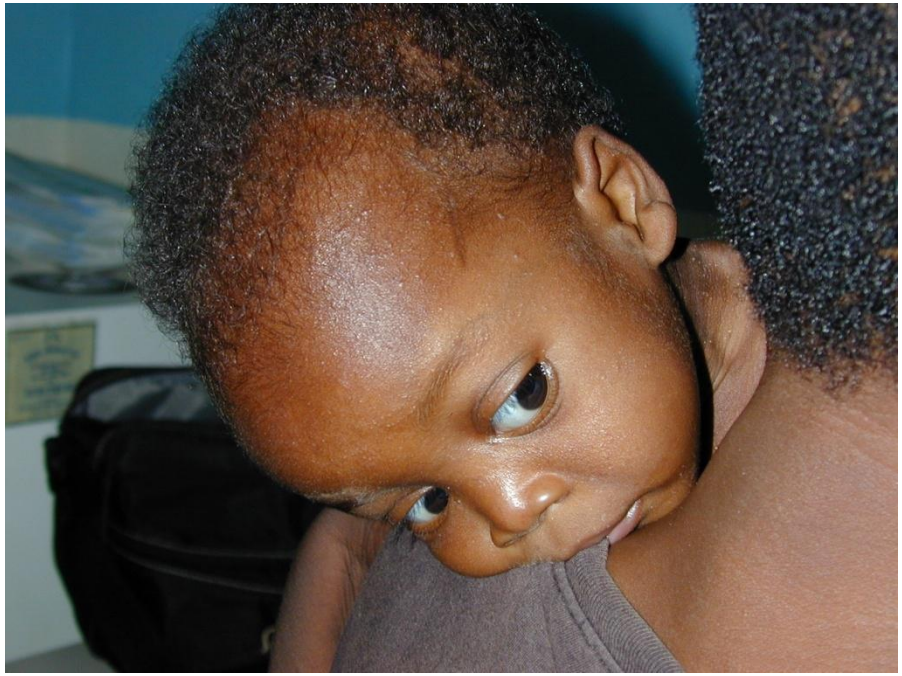
Capillary Refill in Immediate Newborn Period

- Assess centrally over sternum
- Normal range 1 to 3 secs
- Abnormal begins at 4 secs and longer

Pallor



Sunken Eyes



Skin Pinch

Disability

D	AVPU	A	V	P	U
	Can drink / breastfeed?			Y☐	N☐
	Stiff neck			Y☐	N☐
	Bulging fontanelle			Y☐	N☐
If young infant < 2m	Irritable			Y☐	N☐
	Reduced movement / tone			Y☐	N☐

AVPU Scale

- A = Alert
- V = Responds to a voice / sound appropriately
- P = Responds <u>appropriately</u> to pain
- U = Unresponsive / Unconscious

Alert?

Responds to voice

Responds to Pain

Ability to Drink / Breastfeed?

Bulging fontanelle and stiff neck

- Fontanelle should be examined with the infant lying down at rest (i.e. not crying)

General Condition / Nutrition?

General Examination

Thrush **Y**☐ **N**☐ Lymph N > 1cm **Y**☐ **N**☐

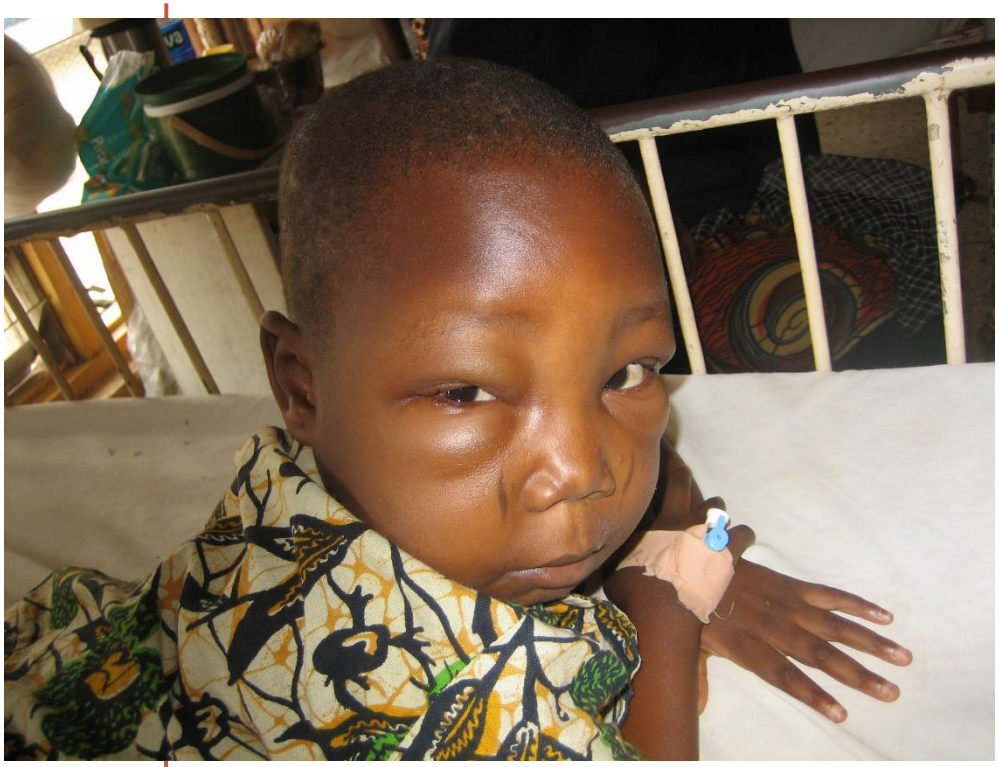
Wrist / Rib signs Rickets **Y**☐ **N**☐

Jaundice	**0**	**+**	**+++**
Oedema	☐None	☐Foot	☐Knee ☐Face

Jaundice

Oedema

Oedema	☐None ☐Foot ☐Knee ☐Face



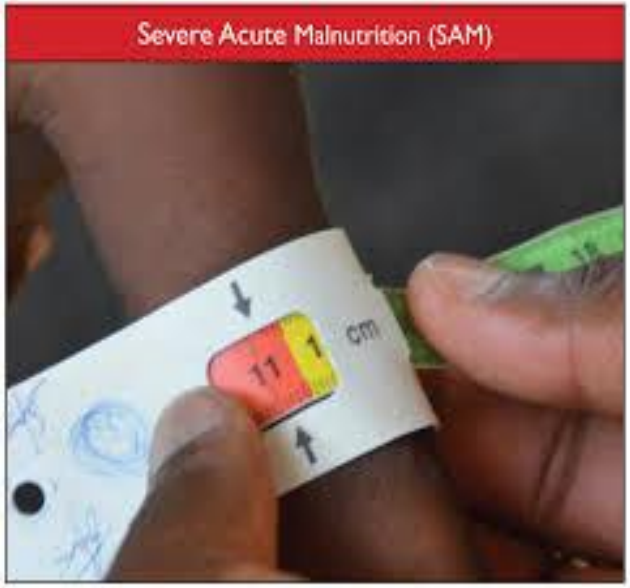
Mid Upper Arm Circumference (MUAC)

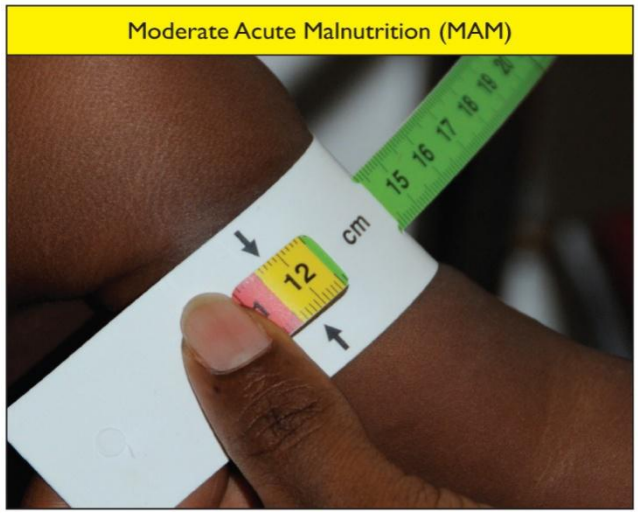
- MUAC is the recommended measure for assessing nutritional status in children

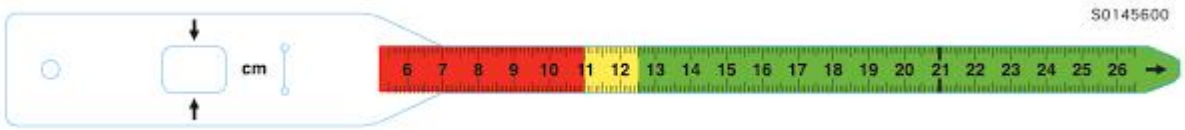
  aged 6 – 59 months

- MUAC is a single linear measurement that does not require arithmetic, table look-up or plotting data on growth charts

- A colour-coded tape is used to determine the level of severity of malnutrition

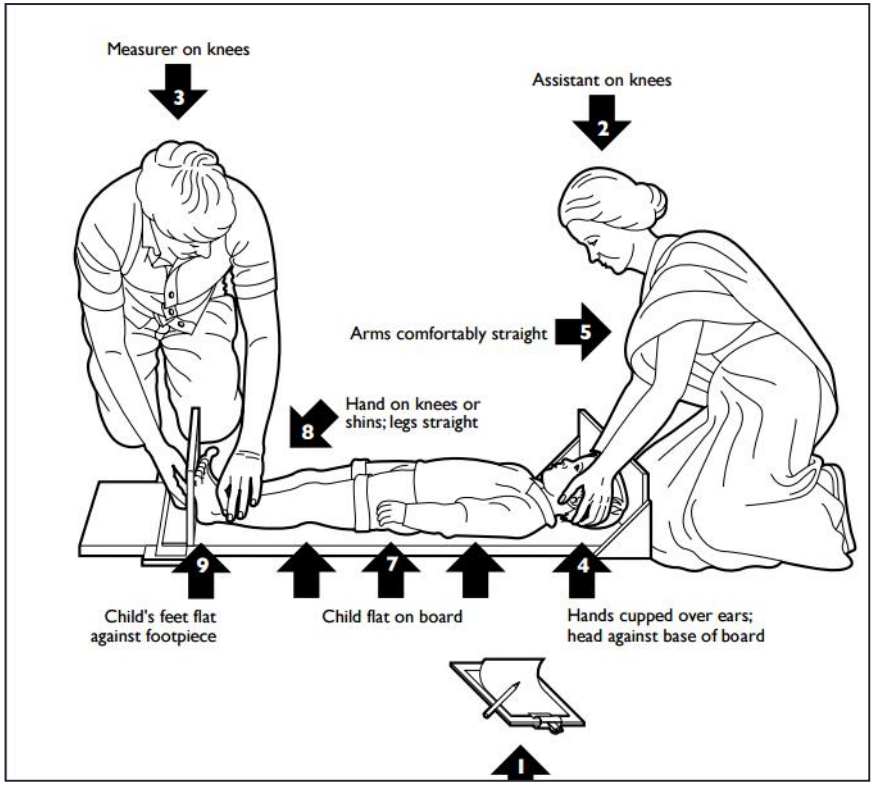






Length measurement

- Children up to 87 cm (or <2 years) are measured while lying down
- Classification of nutritional status based on WHZ score is used in infants <6 months



Definitions of Malnutrition

	MUAC cm (6-59 months)	**WHZ (<6months)**
None	>13.5	>-1
At Risk	12.5 to 13.4	-2 to -1
Moderate	11.5 to 12.4	-3 to -2
Severe	**<11.5**	**<-3**
	Oedema of severe malnutrition	

QUESTIONS?

Summary

- Simple symptoms and signs will help guide basic treatment in 80-90% of children admitted.
- A common approach to interpreting clinical signs helps clinical communication.
- Always be on the look out for additional important signs